

R BASICS

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Outline

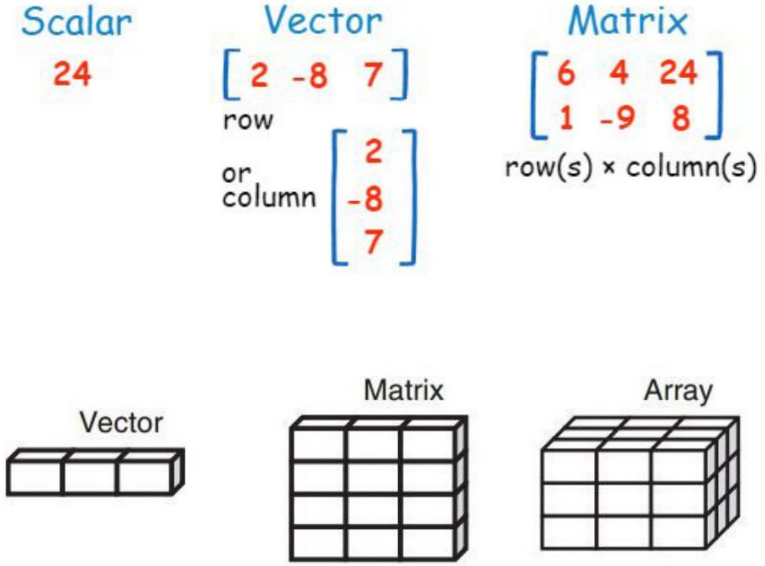
- Data types
- Data structures
- Summary of data structures
- Objects and variables
- Packages
- Missing values
- Mini exercise

Data types

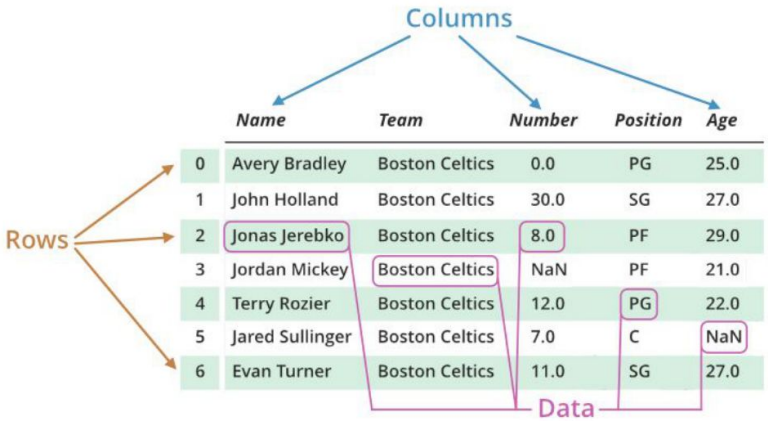
- Character: "hanis", "school"
- Numeric: 1.0, 2.1, 3.0, 3.2
- Integer: 1, 2, 3, 4L
- Logical: TRUE, FALSE
- Complex: e.g. 1+4i
    - Not commonly used
    - Stores a number with an imaginary component

Data structures

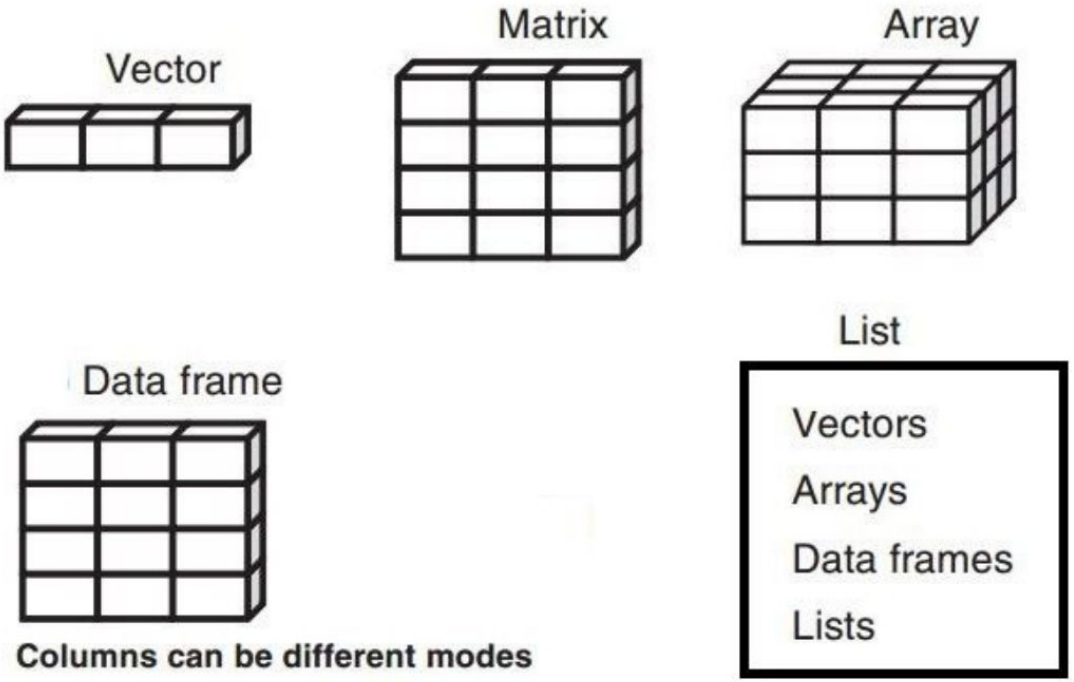
- Most common data structures in R
- Atomic vector: basic data structure; contains one data type at a time
- Matrix
- Array: a matrix with more than one layer; more common in image analysis



- Data frame
  - Variable names in the column names, each observation in a row
  - Each column may contain a different data type
- List
  - May contain different data structures



Summary of data structures



Objects and variables

- Store a value with an assignment operator (<-)
  - `x <- 5`
  - `age <- c(25, 31, 47)`
- Everything in R is an object (number, text, plot, etc)
- Naming: letters, numbers, . and _
  - `age`, `x.1`, `blood_pressure` - valid variable name
  - A good habit - lowercase with underscores (`sbp_mmHg`)
  - Cannot start with a number
- R is case-sensitive: Age and age are not the same

Packages

- Package is a collection of functions built for a specific purpose
  - Example: ggplot2 for plotting
- Meta package is collection of packages (example: tidyverse)
- To use a package:
  - It need to be installed - `install.packages("tidyverse")`
  - It need to be loaded in the session first - `library(tidyverse)`

Missing values

- NA = not available (a real missing value)
- NaN = not a number (e.g. 0/0)
- "NA" in quotes is just text, not missing
- Check with is.na()
- Many functions return NA unless na.rm = TRUE

```
x <- c(5, NA, 9)
is.na(x)
sum(is.na(x))
```

Mini exercise (10 min)

1. Create a vector of 5 ages
2. Create a data frame with id, age, sex (3 rows)
3. Add one NA to the ages
4. Compute the mean of age (with na.rm = TRUE)

Tip: check with str() and class()



Any question?